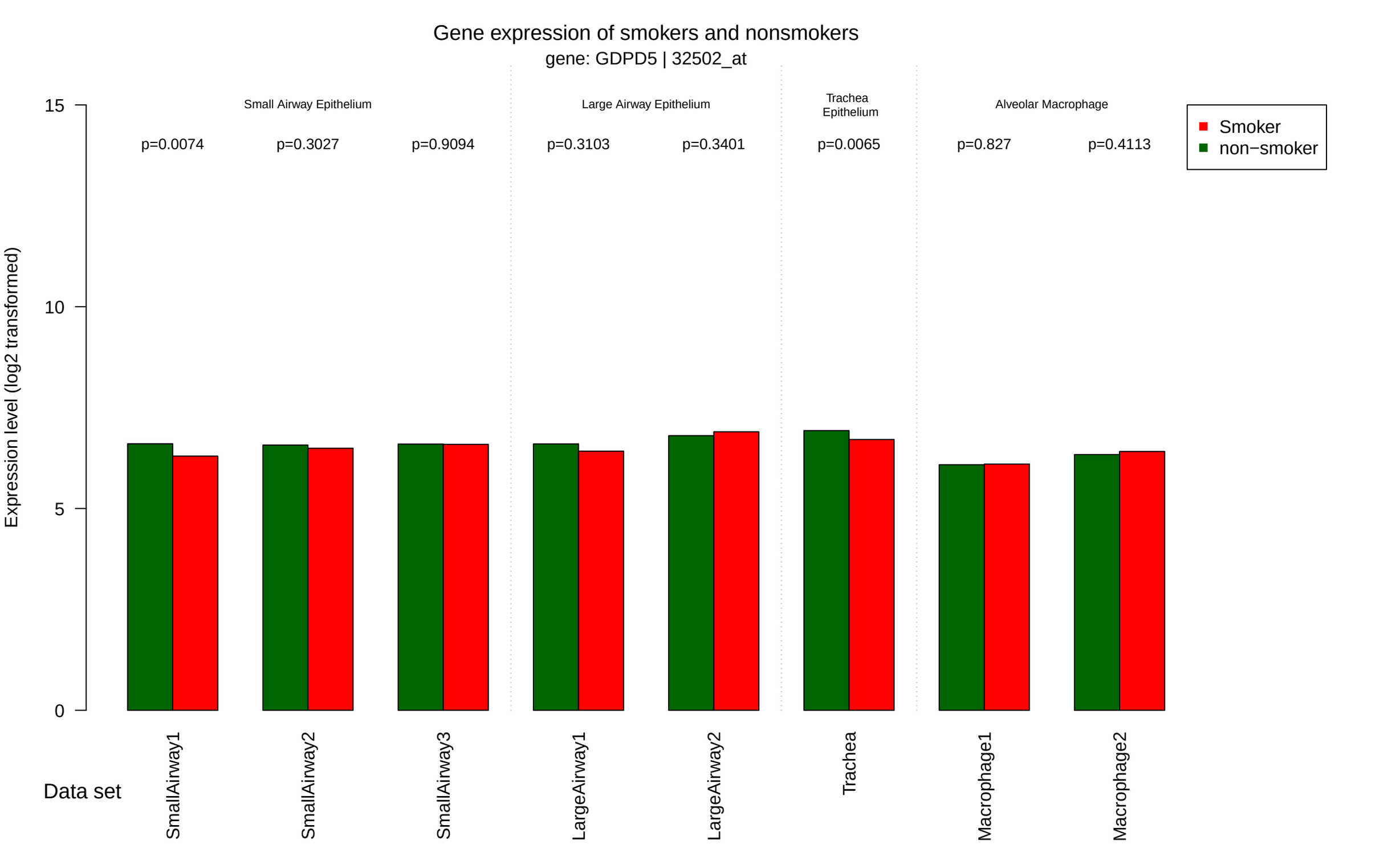
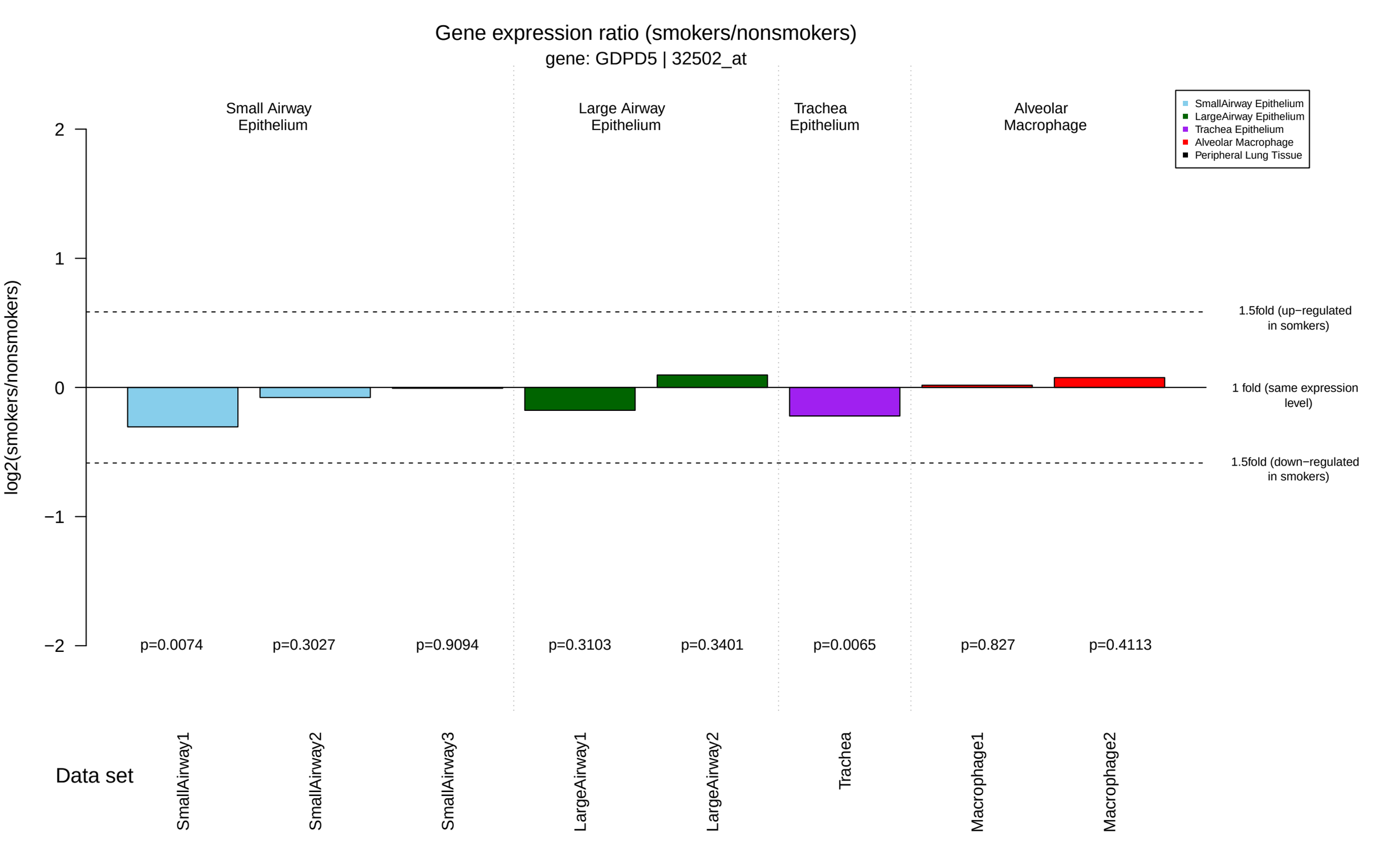
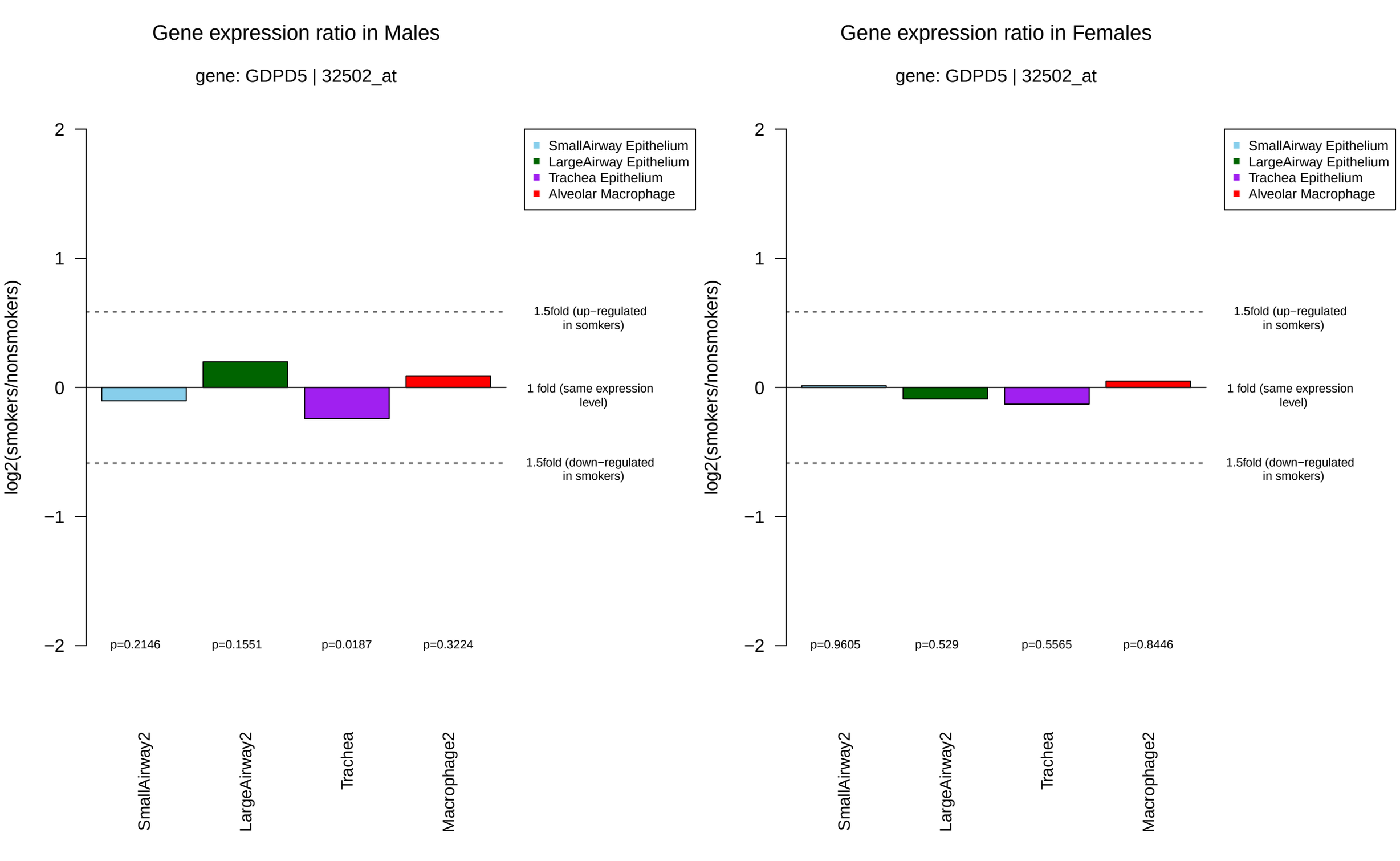
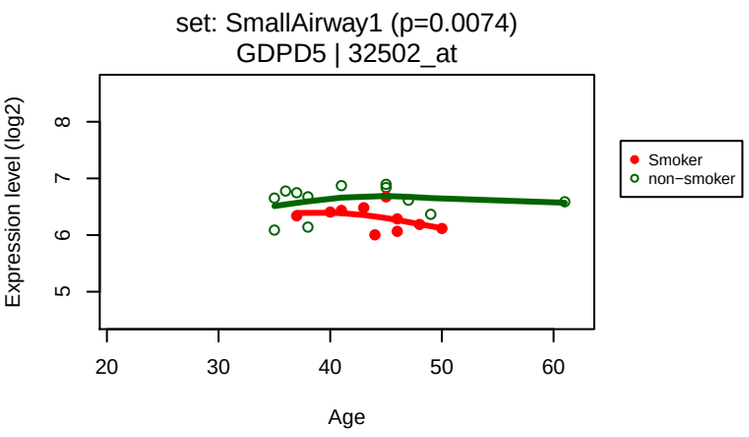
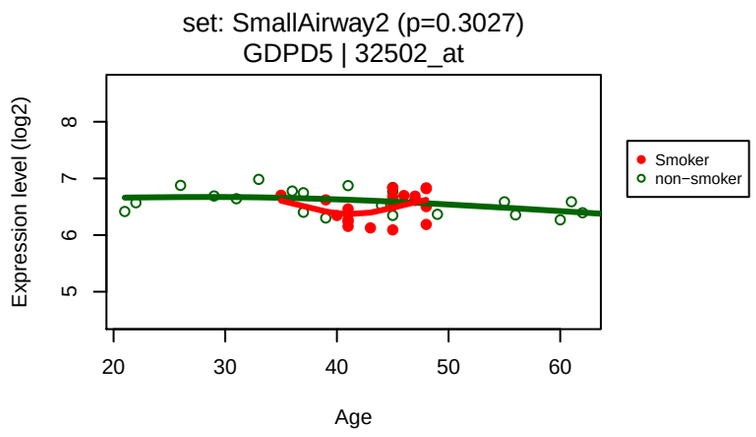
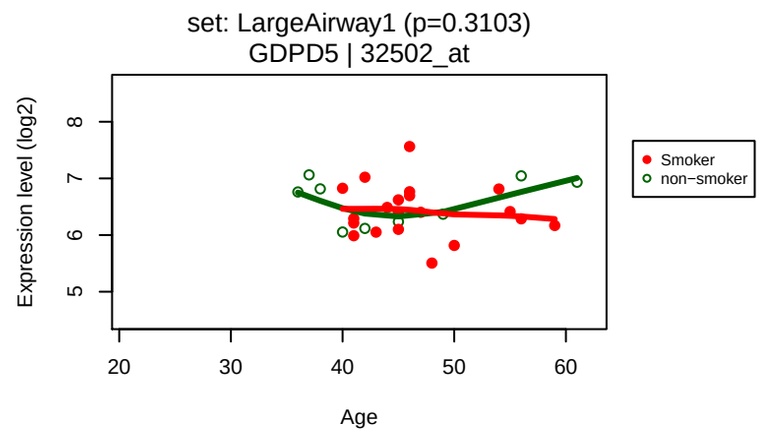

Gene expression of smokers and nonsmokers
gene: GDPD5 | 32502_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0074
p=0.3027
p=0.9094
p=0.3103
p=0.3401
p=0.0065
p=0.827
p=0.4113
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

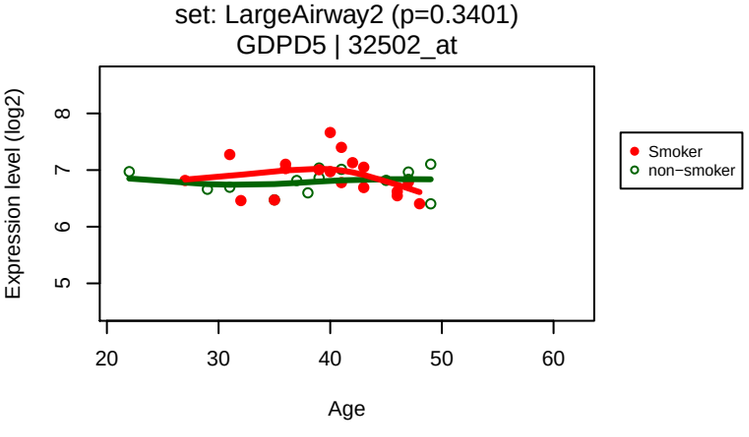

Gene expression ratio (smokers/nonsmokers)
gene: GDPD5 | 32502_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0074
p=0.3027
p=0.9094
p=0.3103
p=0.3401
p=0.0065
p=0.827
p=0.4113
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

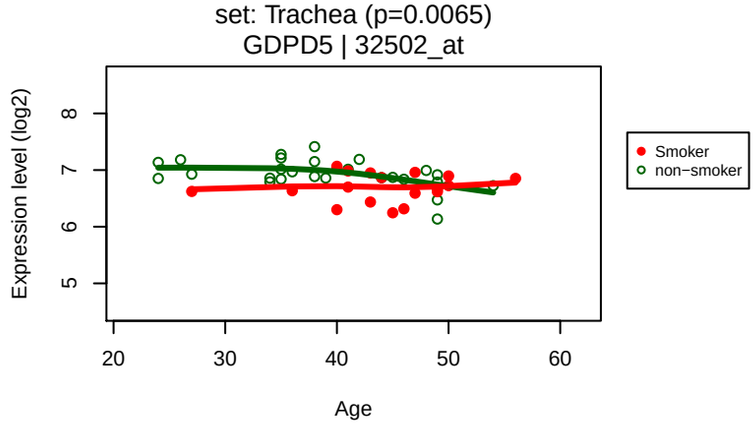

Gene expression ratio in Males
gene: GDPD5 | 32502_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2146
p=0.1551
p=0.0187
p=0.3224
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GDPD5 | 32502_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9605
p=0.529
p=0.5565
p=0.8446
SmallAirway2
LargeAirway2
Trachea
Macrophage2

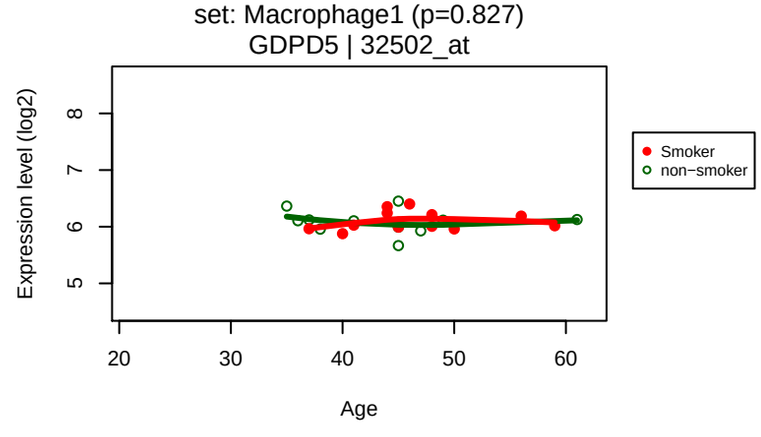

set: SmallAirway1 (p=0.0074)
GDPD5 | 32502_at
Expression level (log2)
Age
Smoker
non−smoker

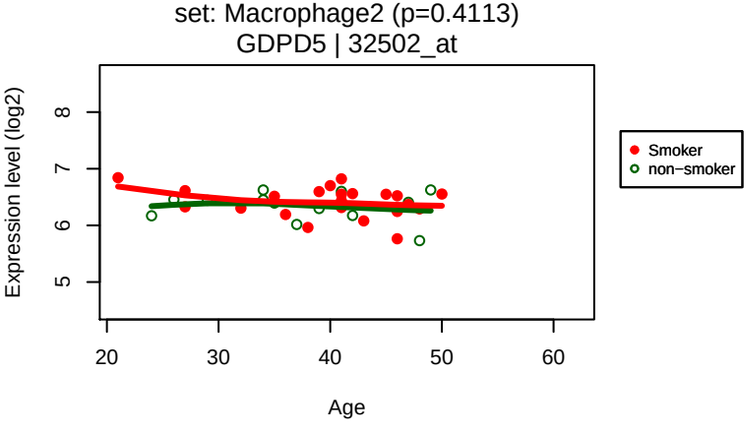

set: SmallAirway2 (p=0.3027)
GDPD5 | 32502_at
Expression level (log2)
Age
Smoker
non−smoker

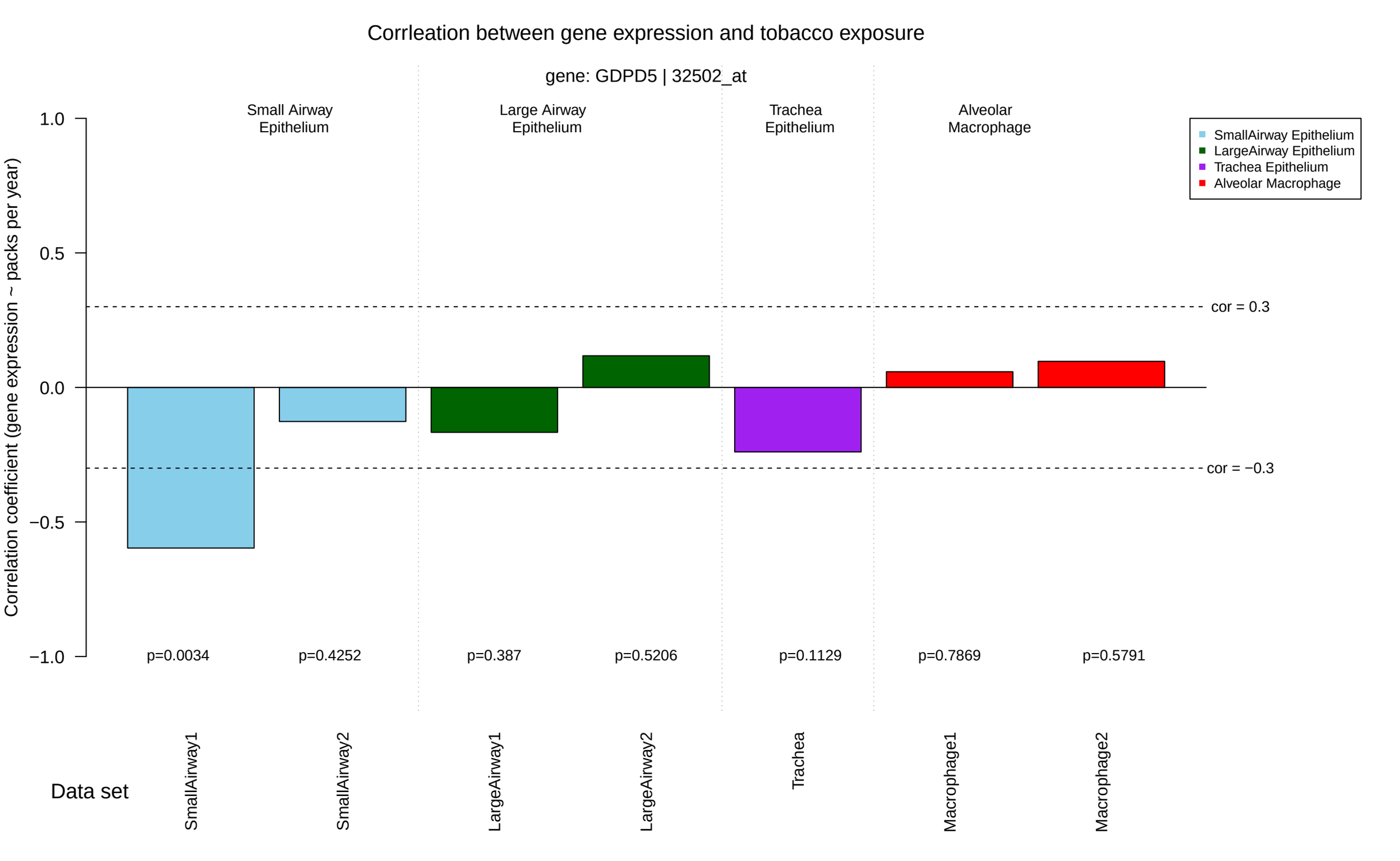

set: LargeAirway1 (p=0.3103)
GDPD5 | 32502_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3401)
GDPD5 | 32502_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0065)
GDPD5 | 32502_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.827)
GDPD5 | 32502_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4113)
GDPD5 | 32502_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GDPD5 | 32502_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0034
p=0.4252
p=0.387
p=0.5206
p=0.1129
p=0.7869
p=0.5791
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2